

PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

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Program/Batch	B082-Networkwalks
Date	17 August 2026
Modules completed	W2-PM1 (Multiple Kali Tools) W2-PM5 (Zenmap Scanning)
Client/Target	1. Networkwalks (secured written permission already) 2. My own local LAN Network
Permission secured from client?	Yes
Phases covered	Phase 1: Reconnaissance & Footprinting Phase 2: Scanning & Network Discovery Phase 3-5: In Progress

1. Liability Disclaimer

I have performed these activities only on the systems & devices where I had secured written permission or the devices/systems that I own myself. All these materials are for education and research purpose only. Do not use anything from here to break the law. The instructor, the authors and Networkwalks are not responsible for what you do with this knowledge. Every action you take is your own responsibility. Misuse can lead to criminal charges, heavy fines, loss of your job and a permanent record. In most countries unauthorised access is a crime even when nothing is damaged.

2. Introduction

This report covers footprinting the networkwalks.com domain using multiple Kali Linux tools (W2-PM1) and scanning my own local network with Zenmap (W2-PM5). One module covers the footprinting phase and the other covers the scanning phase, so together they show how an attacker moves from gathering public information to mapping live hosts on a network. It is the Week 2 part of my ongoing internship program at Networkwalks.

All commands were run in Kali Linux (footprinting) and on a Windows PC with Zenmap installed (scanning). Every step below includes the exact command used, the result I observed, a screenshot as evidence, and a short note on why the finding matters from an attacker's point of view.

3. Tools Used

The table below lists each tool used in this report and its purpose.

Tool	Purpose
Kali Linux & Windows	Operating systems used for reconnaissance activities
WHOIS	Find domain registration details (owner, dates, name servers).
whatweb	Fingerprint web technologies (server, CMS, plugins, IP).
nslookup	Resolve the domain name to its IP address using DNS.
curl -I	Read the HTTP response headers of the website.
wafw00f	Detect whether a Web Application Firewall protects the site.
dnsrecon	Enumerate all DNS records (NS, MX, SPF, TXT, SRV).
Zenmap (Nmap GUI)	Scan the local subnet to find live hosts, IPs and MAC addresses.
Windows CMD	Local IP and MAC address identification

4. Activities Performed

4.1 Footprinting & Reconnaissance

I performed reconnaissance against the networkwalks.com domain using six Kali Linux tools: **WHOIS**, **WhatWeb**, **Nslookup**, **Curl**, **Wafw00f** and **DNSRecon**. Each tool was used to collect a different type of information about the target.

First, I used **WHOIS** to obtain publicly available domain registration information and identify the domain's name servers. The results provided information about the domain registration and hosting infrastructure.

I then used **WhatWeb** to identify technologies used by the website. The results identified **WordPress 7.0.4** and **WP Download Manager 3.3.58**, along with other information exposed by the website.

Using **Nslookup**, I resolved the domain name to its IP address. The provided result identified **192.232.216.135**.

I used **Curl** with the `-I` option to inspect the HTTP response headers. This provided additional information about the web application and exposed the WordPress REST API endpoint `/wp-json/`.

Next, I used **Wafw00f** to determine whether a Web Application Firewall was protecting the website. The result identified **ModSecurity (SpiderLabs)**.

Finally, I used **DNSRecon** to enumerate DNS records. The results provided information relating to name servers, mail servers, SPF/TXT records, service records and DNS software information.

4.1.1 WHOIS Analysis

command: `whois networkwalks.com`

Domain Created: 2019-11-06

Expiry Date: 2027-11-06

Registrar: GoDaddy.com, LLC

Registrant: Domains By Proxy, LLC (Privacy Protection)

Name Servers: NS6135.HOSTGATOR.COM, NS6136.HOSTGATOR.COM,
NS29.DOMAINCONTROL.COM, NS30.DOMAINCONTROL.COM

DNSSEC: unsigned

Status: clientDeleteProhibited, clientRenewProhibited, clientTransferProhibited, clientUpdateProhibited

Analysis:

The domain has been registered since 2019 and will expire in 2027. Privacy protection is enabled through Domains By Proxy, which hides the actual registrant's contact information. The domain is locked against unauthorized transfers and updates, which is a positive security measure.

4.1.2 WhatWeb Analysis

command: `whois networkwalks.com`

Domain Created: 2019-11-06

Expiry Date: 2027-11-06

Registrar: GoDaddy.com, LLC

Registrant: Domains By Proxy, LLC (Privacy Protection)

Name Servers: NS6135.HOSTGATOR.COM, NS6136.HOSTGATOR.COM,
NS29.DOMAINCONTROL.COM, NS30.DOMAINCONTROL.COM

DNSSEC: unsigned

Status: clientDeleteProhibited, clientRenewProhibited, clientTransferProhibited, clientUpdateProhibited

Analysis:

The domain has been registered since 2019 and will expire in 2027. Privacy protection is enabled through Domains By Proxy, which hides the actual registrant's contact information. The domain is locked against unauthorized transfers and updates, which is a positive security measure.

4.1.3 Nslookup Analysis

command: nslookup networkwalks.com

Server: 192.168.0.1
Address: 192.168.0.1#53

Non-authoritative answer:
Name: networkwalks.com
Address: 192.232.216.135

Analysis:

The domain resolves to IP address 192.232.216.135. The DNS query was handled by the local router (192.168.0.1) which forwarded the request to upstream DNS servers.

The IP address reveals the server's location and can be used for further reconnaissance, including port scanning and service enumeration.

4.1.4 Curl HTTP Header Analysis

command: curl -I networkwalks.com

HTTP/1.1 301 Moved Permanently
Date: Thu, 20 Aug 2026 14:01:45 GMT
Server: Apache
Permissions-Policy: private-state-token-redemption=(self "https://www.google.com"
"https://www.gstatic.com" "https://recaptcha.net"
"https://challenges.cloudflare.com" "https://hcaptcha.com")
Expires: Thu, 20 Aug 2026 15:01:45 GMT
Cache-Control: max-age=3600
X-Redirect-By: WordPress - Really Simple Security
Set-Cookie: __wpdm_client=8780f3d9d328def1ddfe431769d6342a; path=/;
domain=networkwalks.com; HttpOnly
Upgrade: h2,h2c
Connection: Upgrade
Location: https://networkwalks.com/
Referrer-Policy: no-referrer-when-downgrade
X-Endurance-Cache-Level: 0
X-nginx-cache: WordPress
Content-Type: text/html; charset=UTF-8

Analysis:

The site performs a 301 redirect from HTTP to HTTPS, which is good security practice. The X-Redirect-By header reveals the "Really Simple Security" WordPress plugin is handling the redirect. The cookie __wpdm_client is set with HttpOnly flag, which provides some protection against XSS attacks. The X-nginx-cache header reveals Nginx caching is used.

4.1.5 Wafw00f Analysis

[*] Checking https://networkwalks.com
[+] The site https://networkwalks.com is behind ModSecurity (SpiderLabs) WAF.
[~] Number of requests: 2

Analysis:

The website is protected by ModSecurity Web Application Firewall from SpiderLabs. The WAF was identified after only 2 requests.

4.1.6 DNSRecon Analysis

command: dnsrecon -d networkwalks.com

```
[*] SOA ns6135.hostgator.com 50.87.144.87
[*] NS ns6135.hostgator.com 50.87.144.87
[*] Bind Version for 50.87.144.87 "9.16.23-RH"
[*] NS ns6136.hostgator.com 192.232.216.131
[*] MX mail.networkwalks.com 192.232.216.135
[*] A networkwalks.com 192.232.216.135
[*] SRV _autodiscover._tcp.networkwalks.com cpanel.emaildiscovery.cpanel.net
    184.94.204.9 443
[*] SRV _autodiscover._tcp.networkwalks.com cpanel.emaildiscovery.cpanel.net
    184.94.203.9 443
[*] SRV _autodiscover._tcp.networkwalks.com cpanel.emaildiscovery.cpanel.net
    184.94.204.11 443
[*] SRV _autodiscover._tcp.networkwalks.com cpanel.emaildiscovery.cpanel.net
    184.94.203.14 443
[*] SRV _autodiscover._tcp.networkwalks.com cpanel.emaildiscovery.cpanel.net
    184.94.204.15 443
[*] SRV _autodiscover._tcp.networkwalks.com cpanel.emaildiscovery.cpanel.net
    184.94.203.11 443
[*] SRV _autodiscover._tcp.networkwalks.com cpanel.emaildiscovery.cpanel.net
    184.94.203.15 443
[*] SRV _autodiscover._tcp.networkwalks.com cpanel.emaildiscovery.cpanel.net
    184.94.204.14 443
[*] 8 Records Found
```

Analysis:

The DNS records reveal extensive infrastructure information. The BIND version is explicitly exposed, which is a security concern. The MX record points to the same IP as the web server, indicating the email and web services are hosted on the same server. The SRV records show the site uses cPanel for email services.

4.2.1 Local Network Configuration

command: ifconfig

```
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.0.11 netmask 255.255.255.0 broadcast 192.168.0.255
    inet6 2001:e68:5450:1561:5ff1:6f1:fc7b:1684 prefixlen 64 scopeid 0x0<global>
    inet6 2001:e68:5450:1561:20c:29ff:fe12:21ff prefixlen 64 scopeid 0x0<global>
    inet6 fe80::20c:29ff:fe12:21ff prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:12:21:ff txqueuelen 1000 (Ethernet)
    RX packets 30512 bytes 31955558 (30.4 MiB)
    RX errors 18 dropped 19 overruns 0 frame 0
    TX packets 14011 bytes 2672087 (2.5 MiB)
```

TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
device interrupt 19 base 0x2000

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
inet 127.0.0.1 netmask 255.0.0.0
inet6 ::1 prefixlen 128 scopeid 0x10<host>
loop txqueuelen 1000 (Local Loopback)
RX packets 10 bytes 580 (580.0 B)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 10 bytes 580 (580.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

Findings:

IP Address: 192.168.0.11

Subnet Mask: 255.255.255.0 (/24)

Broadcast Address: 192.168.0.255

MAC Address: 00:0c:29:12:21:ff

Network Interface: eth0 (Ethernet)

Analysis:

The Kali machine is on the 192.168.0.0/24 subnet with IP address 192.168.0.11. The subnet mask of 255.255.255.0 means the network can support up to 254 hosts (192.168.0.1 to 192.168.0.254).

4.2 Network Scanning with Zenmap

For the second activity, I used Nmap to perform network discovery on my local network. The practical required me to identify my local IP address and subnet, discover live hosts, and identify their IP and MAC addresses.

Nmap scan report for 192.168.0.1 (192.168.0.1)

Host is up (0.0025s latency).

MAC Address: A4:2A:95:B2:A4:62 (D-Link International)

Nmap scan report for ty_wr (192.168.0.2)

Host is up (0.097s latency).

MAC Address: 70:89:76:CE:C5:0A (Tuya Smart)

Nmap scan report for redmi-13c (192.168.0.3)

Host is up (0.11s latency).

MAC Address: 9A:46:50:31:B3:EF (Unknown)

Nmap scan report for mitv (192.168.0.4)

Host is up (0.083s latency).

MAC Address: 50:E4:78:69:4A:3F (Sichuan AI-Link Technology)

Nmap scan report for redmi-15-5g (192.168.0.5)

Host is up (0.097s latency).

MAC Address: CA:3E:72:26:E1:1B (Unknown)

Nmap scan report for 192.168.0.6 (192.168.0.6)

Host is up (0.00018s latency).

MAC Address: 70:08:94:2B:C1:AB (Liteon Technology)

Nmap scan report for honor-x9c (192.168.0.7)

Host is up (0.11s latency).

MAC Address: 24:AE:CC:3D:E1:FB (Honor Device)

Nmap scan report for win-9sulpohlo0j (192.168.0.9)

Host is up (0.00017s latency).

MAC Address: 00:0C:29:E6:FA:6B (VMware)

Nmap scan report for laptop-3nbcjkco (192.168.0.11)

Host is up.

Nmap done: 256 IP addresses (9 hosts up) scanned in 4.07 seconds

After completing the scan, I opened the **Topology** section in Zenmap, enabled the legend and saved the network topology in PDF format as required by the practical task.

Note: The actual subnet, number of hosts and addresses should be replaced with the results from my own network when submitting the report.

5. Risk Analysis / Impact

Based on the information collected during the footprinting and network scanning activities, I identified the following potential risks.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Multiple IoT devices on network	Tuya Smart, Xiaomi TV discovered	IoT devices often have poor security and can be used as entry points	● Medium
2	Router manufacturer identified	D-Link International (192.168.0.1)	Consumer routers may have known vulnerabilities	● Medium
3	Random MAC addresses detected	Redmi 13C, Redmi 15 5G (locally administered MACs)	Privacy feature used, but devices still need to be secured	● Low
4	Multiple mobile devices	3 smartphones detected	Mobile devices may contain sensitive data	● Low
5	VMware virtual machines	Two VMs detected (Windows and Kali)	VMs may have vulnerable services	● Medium
6	Hostname exposure	Hostnames reveal device types (redmi-13c, mitv, honor-x9c)	Helps attackers identify devices for targeted attacks	● Low
7	cPanel Control Panel Exposed	Port 2083 open on networkwalks.com	cPanel provides administrative access to the entire hosting account, including email, databases, DNS, and file management	● Critical
8	Recent cPanel Critical Vulnerability	CVE-2026-41940 - Authentication bypass in cPanel/WHM	Unauthenticated attackers can gain full control panel access without credentials	● Critical

Risk level key: ● Critical ● Medium ● Low

The risks above are observations from the footprinting and scanning exercises, not confirmed vulnerabilities.

The practical exercises primarily involved information gathering and host discovery. No exploitation or vulnerability validation was performed as part of these two modules.

Therefore, the presence of information such as a software version, IP address or DNS record does not by itself mean that the system is vulnerable. Further authorized security testing would be required to confirm any actual vulnerability.

6. Recommendations

Based on the observations from these activities, I recommend the following security improvements:

1. **Review publicly exposed technology information**
Organizations should regularly review what information about their web technologies, CMS and plugins is publicly visible.
2. **Keep software updated**
CMS platforms, plugins and other web technologies should be regularly updated and reviewed against current security advisories.
3. **Review HTTP headers**
HTTP response headers should be reviewed to determine whether unnecessary technical information is being exposed.
4. **Review DNS records regularly**
DNS records should be checked periodically to ensure that only required information and services are publicly exposed.
5. **Properly configure and monitor the WAF**
Keep the WAF (ModSecurity) enabled and tuned, since it already blocks naive attacks.
6. **Perform regular internal network discovery**
Organizations should periodically scan their own networks to identify active devices.
7. **Investigate unknown devices**
Any unexpected device discovered during network scanning should be investigated and verified.
8. **Maintain network documentation**
Network topology and device information should be documented and updated regularly.
9. **Perform security testing with authorization**
Reconnaissance and scanning should only be performed against systems and networks where appropriate authorization has been provided.

7. Conclusion

During Week 2 of my Cybersecurity & Ethical Hacking internship, I completed practical activities covering footprinting, reconnaissance and network scanning.

The exercises showed me that information gathering is an important part of cybersecurity. Even before attempting to exploit a system, a security professional can learn a significant amount about an environment by carefully analyzing publicly available information and network responses. The discovery of an exposed cPanel control panel with a critical known vulnerability demonstrates how passive reconnaissance can identify high-risk attack vectors.

I also learned that technical findings should be documented clearly. A good cybersecurity report should explain what was performed, what was discovered, what the observation means, what risk it may create, and what can be done to reduce that risk.

Finally, I learned that reconnaissance and scanning must always be performed within an authorized scope. These activities were completed as part of the assigned educational cybersecurity lab.

8. Evidences Collected

```
(kali@kali: ~)
$ whois networkwalks.com
Domain Name: NETWORKWALKS.COM
Registry domain ID: 2452319255_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.godaddy.com
Registrar URL: http://www.godaddy.com
Updated Date: 2025-11-12T18:08:42Z
Creation Date: 2019-11-06T22:51:46Z
Registry Expiry Date: 2027-11-06T22:51:46Z
Registrar: GoDaddy.com, LLC
Registrar IANA ID: 146
Registrar Abuse Contact Email: abuse@godaddy.com
Registrar Abuse Contact Phone: 480-624-2505
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeletePr
hibited
Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProh
ibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransf
erProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdatePr
hibited
Name Server: NS6135.HOSTGATOR.COM
Name Server: NS6136.HOSTGATOR.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wi
cf/
>>> Last update of whois database: 2026-08-20T14:00:31Z <<<

For more information on Whois status codes, please visit https://icann.org/ep
p

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiratio
n
date of the domain name registrant's agreement with the sponsoring
registrar. Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.

TERMS OF USE: You are not authorized to access or query our Whois
database through the use of electronic processes that are high-volume and
automated except as reasonably necessary to register domain names or
modify existing registrations; the Data in VeriSign Global Registry
Services' ("VeriSign") Whois database is provided by VeriSign for
information purposes only, and to assist persons in obtaining information
about or related to a domain name registration record. VeriSign does not
guarantee its accuracy. By submitting a Whois query, you agree to abide
by the following terms of use: You agree that you may use this Data only
```

cPanel

Username

Remember me

Password

Reset Password

English German Danish Estonian English Español Español Latinoamérica Français

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Domain Status: clientUpdateProhibited <https://icann.org/epp#clientUpdateProhibited>
 Domain Status: clientRenewProhibited <https://icann.org/epp#clientRenewProhibited>
 Domain Status: clientDeleteProhibited <https://icann.org/epp#clientDeleteProhibited>
 Registry Registrant ID: Not Available From Registry
 Registrant Name: Registration Private
 Registrant Organization: Domains By Proxy, LLC
 Registrant Street: DomainsByProxy.com
 Registrant Street: 100 S. Mill Ave, Suite 1600
 Registrant City: Tempe
 Registrant State/Province: Arizona
 Registrant Postal Code: 85281
 Registrant Country: US
 Registrant Phone: +1.4806242599
 Registrant Phone Ext:
 Registrant Fax:
 Registrant Fax Ext:
 Registrant Email: <https://www.godaddy.com/whois/results.aspx?domain=networkwalks.com&action=contactDomainOwner>
 Registry Tech ID: Not Available From Registry
 Tech Name: Registration Private
 Tech Organization: Domains By Proxy, LLC
 Tech Street: DomainsByProxy.com
 Tech Street: 100 S. Mill Ave, Suite 1600
 Tech City: Tempe
 Tech State/Province: Arizona
 Tech Postal Code: 85281
 Tech Country: US
 Tech Phone: +1.4806242599
 Tech Phone Ext:
 Tech Fax:
 Tech Fax Ext:
 Tech Email: <https://www.godaddy.com/whois/results.aspx?domain=networkwalks.com&action=contactDomainOwner>
 Name Server: NS6135.HOSTGATOR.COM
 Name Server: NS6136.HOSTGATOR.COM
 Name Server: NS29.DOMAINCONTROL.COM
 Name Server: NS30.DOMAINCONTROL.COM
 DNSSEC: unsigned
 URL of the ICANN WHOIS Data Problem Reporting System: <http://wdprs.internic.net/>
 >>> Last update of WHOIS database: 2026-08-20T14:00:49Z <<<
 For more information on Whois status codes, please visit <https://icann.org/epp>
 p

```
(kali㉿kali)-[~]  
└─$ nslookup networkwalks.com  
Server:          192.168.0.1  
Address:         192.168.0.1#53  
  
Non-authoritative answer:  
Name:   networkwalks.com  
Address: 192.232.216.135
```

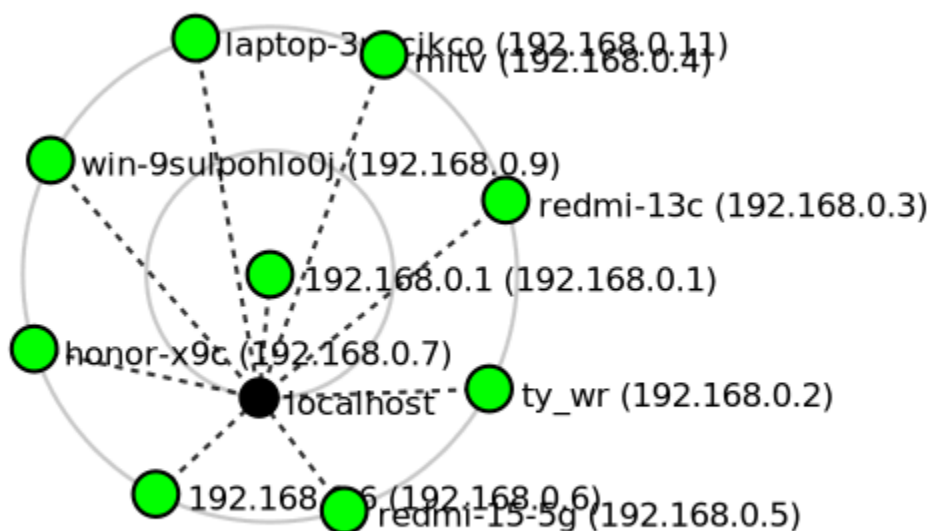
[illegible]

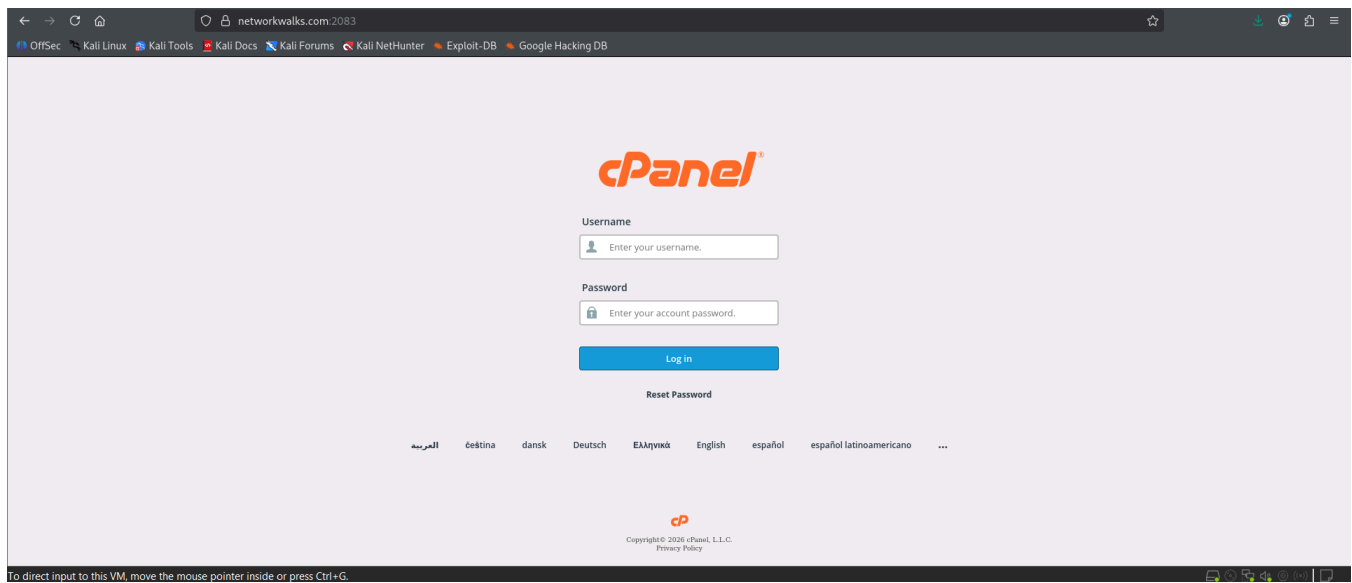
Target: 192.168.0.0/24 Profile: Pingscan

Command: nmap -sn 192.168.0.0/24

OS	Host	Details
192.168.0.1	ty_wr (192.168.0.1)	Starting Nmap 7.99 (https://nmap.org) at 2026-08-20 10:32 -0400 Nmap scan report for 192.168.0.1 (192.168.0.1) Host is up (0.0019s latency).
192.168.0.2	redmi-13c (192.168.0.2)	MAC Address: A4:2A:95:B2:A4:62 (D-Link International) Nmap scan report for ty_wr (192.168.0.2) Host is up (0.064s latency).
192.168.0.3	redmi-15-5g (192.168.0.3)	MAC Address: 70:89:76:CE:C5:0A (Tuya Smart) Nmap scan report for redmi-13c (192.168.0.3) Host is up (0.064s latency).
192.168.0.4	honor-x9c (192.168.0.4)	MAC Address: 9A:46:50:31:B3:EF (Unknown) Nmap scan report for mitv (192.168.0.4) Host is up (0.048s latency).
192.168.0.5	win-9sulpohlo0j (192.168.0.5)	MAC Address: 50:E4:78:69:4A:3F (Sichuan AI-Link Technology) Nmap scan report for redmi-15-5g (192.168.0.5) Host is up (0.064s latency).
192.168.0.6	192.168.0.6 (192.168.0.6)	MAC Address: CA:3E:72:26:E1:1B (Unknown) Nmap scan report for 192.168.0.6 (192.168.0.6) Host is up (0.00020s latency).
192.168.0.7	redmi-13c (192.168.0.7)	MAC Address: 70:08:94:2B:C1:AB (Liteon Technology) Nmap scan report for honor-x9c (192.168.0.7) Host is up (0.076s latency).
192.168.0.9	ty_wr (192.168.0.9)	MAC Address: 24:AE:CC:3D:E1:FB (Honor Device) Nmap scan report for win-9sulpohlo0j (192.168.0.9) Host is up (0.00032s latency).
192.168.0.11	laptop-3nbcjkco (192.168.0.11)	MAC Address: 00:0C:29:E6:FA:6B (VMware) Nmap scan report for laptop-3nbcjkco (192.168.0.11) Host is up.

Nmap done: 256 IP addresses (9 hosts up) scanned in 2.84 seconds





-End-

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 **Project Information**
Program Name: Cybersecurity program at Networkwalks | **Week:** 02 | **Repository:** GitHub